

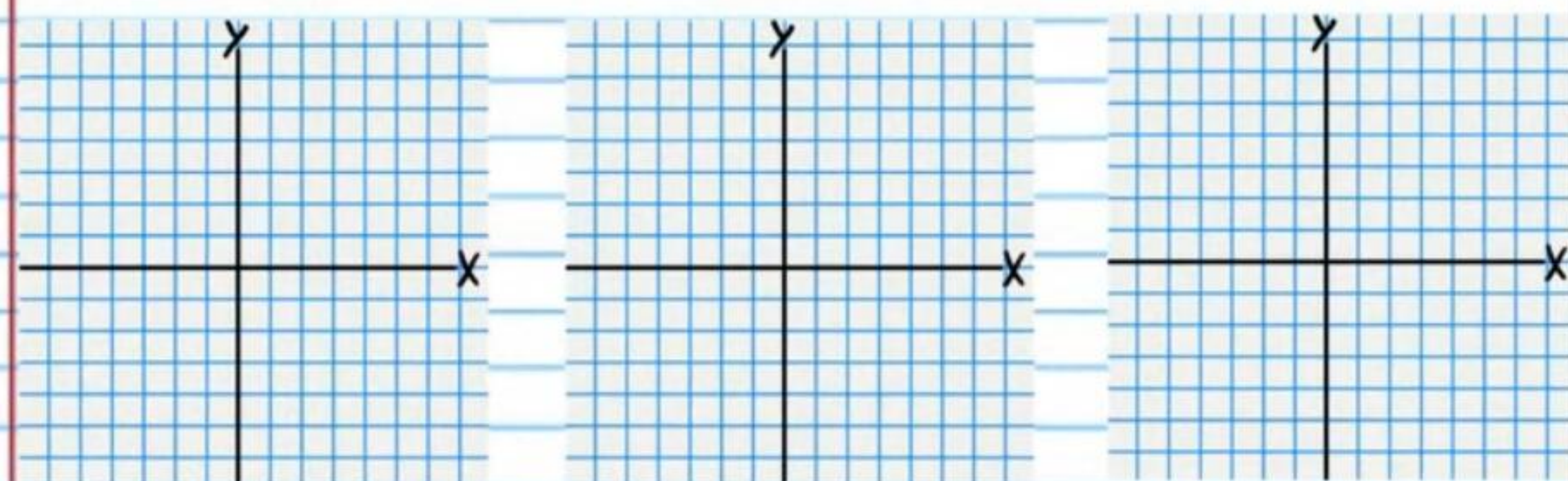
Transformations and the Library of Relations (Part VI) Homework

Change the form of each equation below so that it can be graphed with the transformation method. Draw the parent graph. Then draw each transformation, and write the equation for each graph. Draw any asymptotes as dotted lines.

① $y = \frac{\sqrt{x}}{3} - 1$

Altered Form:

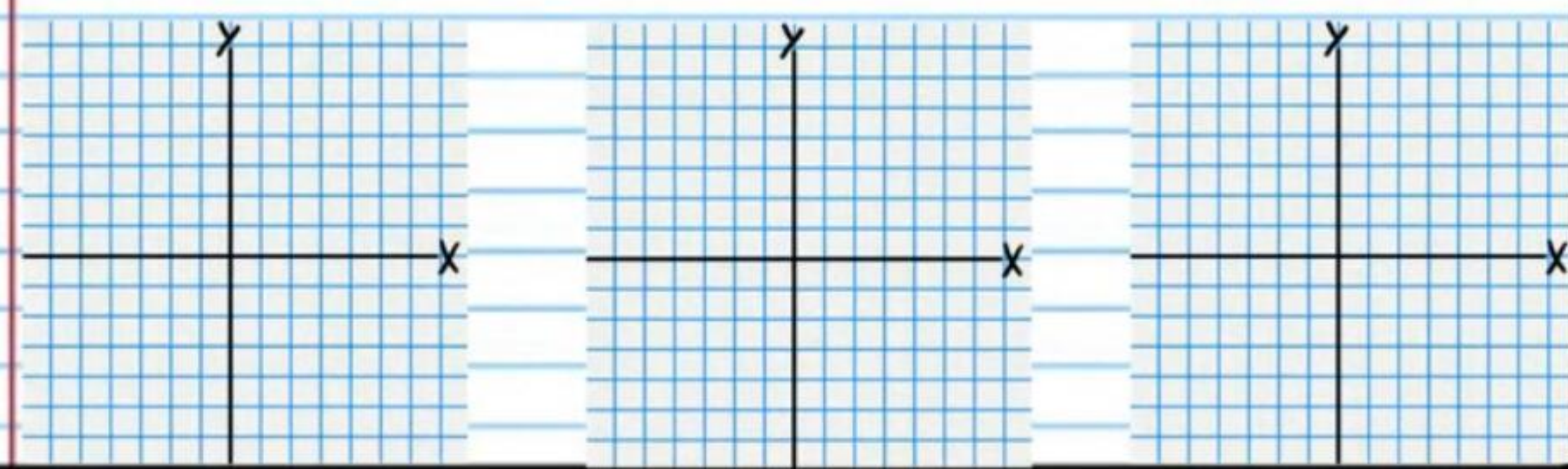
Parent Relation Transformation #1 Transformation #2



② $y = 2 \log_3 x + 2$

Altered Form:

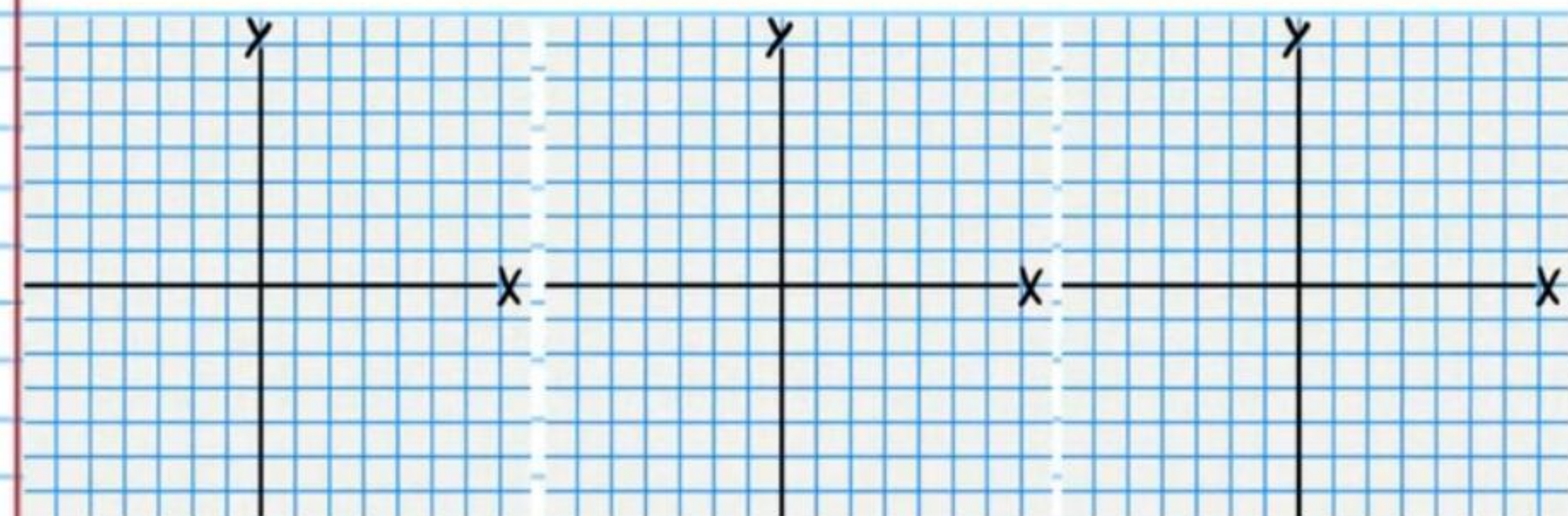
Parent Relation Transformation #1 Transformation #2



③ $y = -|x| + 5$

Altered Form:

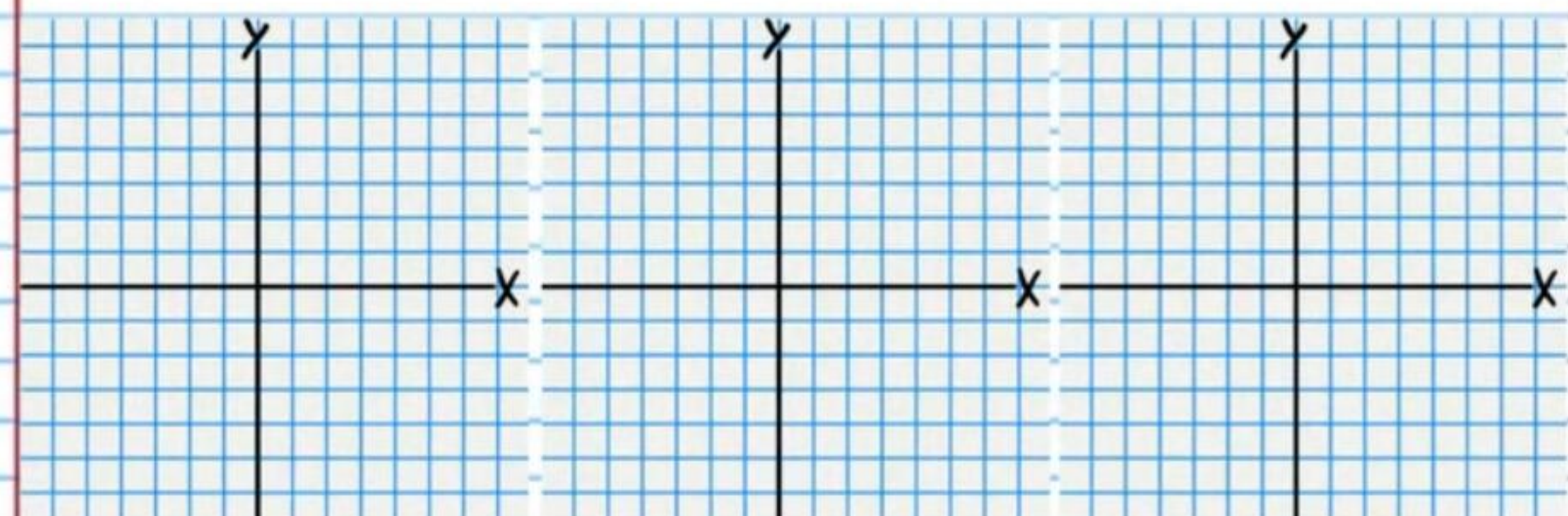
Parent Relation Transformation #1 Transformation #2



④ $y = -\log_2 x - 3$

Altered Form:

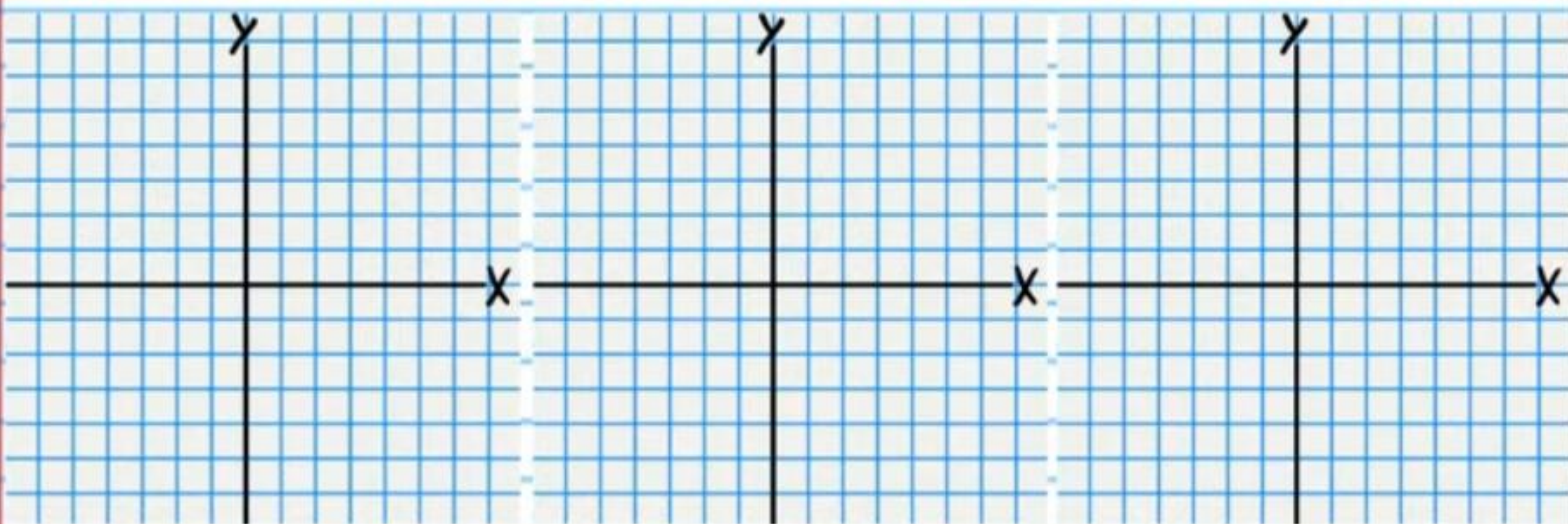
Parent Relation Transformation #1 Transformation #2



$$\textcircled{5} y = 2(x+1)^2 - 4$$

Altered Form:

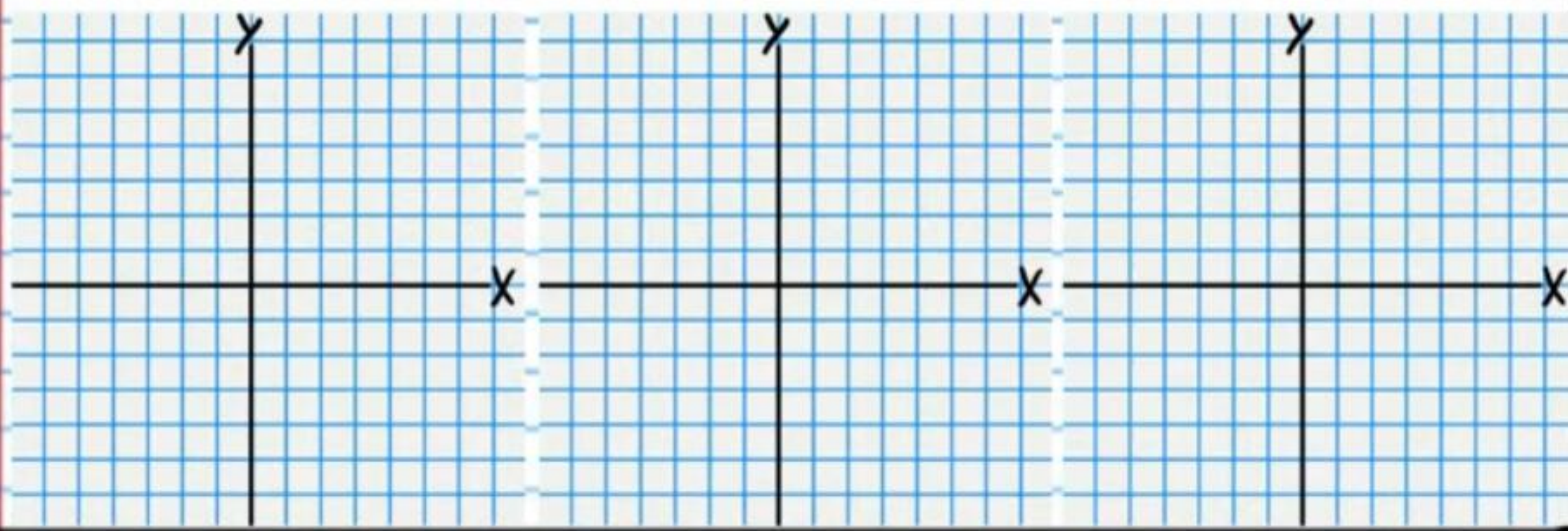
Parent Relation Transformation #1 Transformation #2



$$\textcircled{6} y = -\frac{1}{x-2} + 1$$

Altered Form:

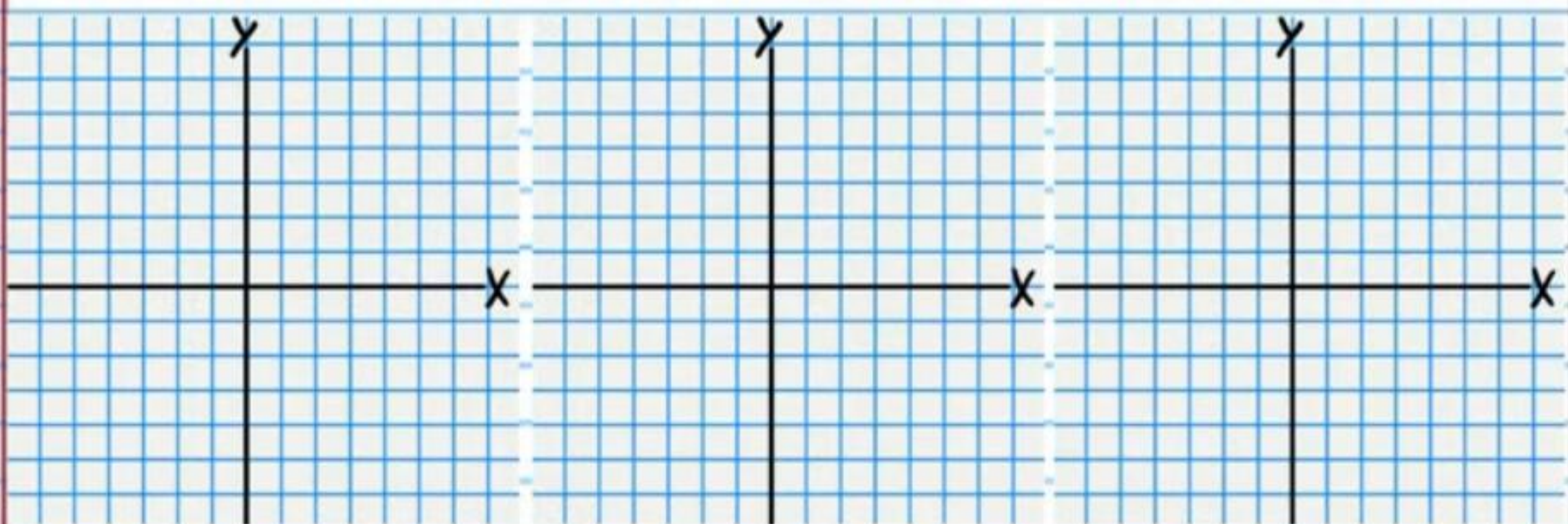
Parent Relation Transformation #1 Transformation #2



$$\textcircled{7} y = \frac{2}{x+3} - 3$$

Altered Form:

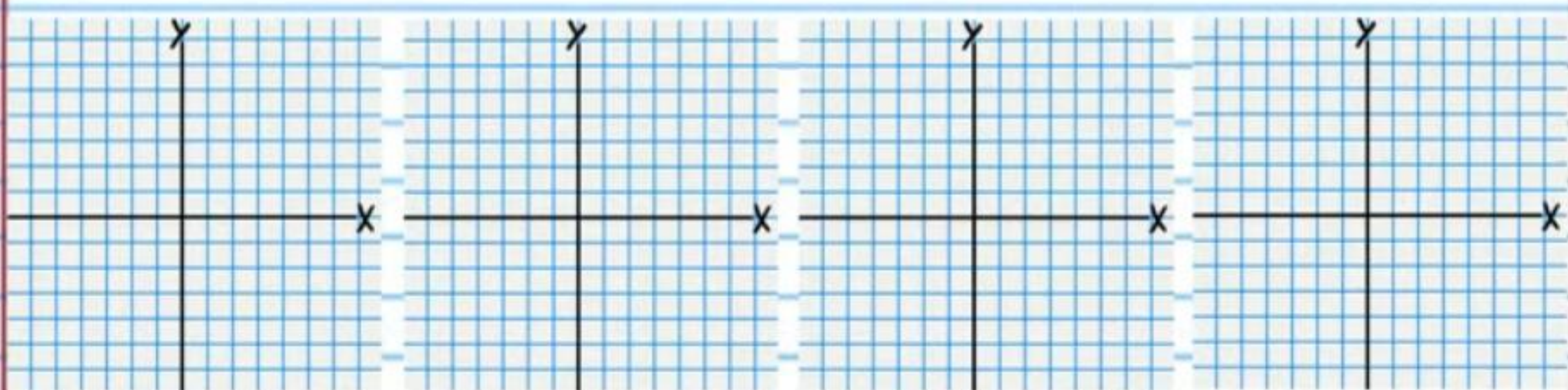
Parent Relation Transformation #1 Transformation #2



$$\textcircled{8} y = -\left|\frac{x}{2} - 3\right| + 3$$

Altered Form:

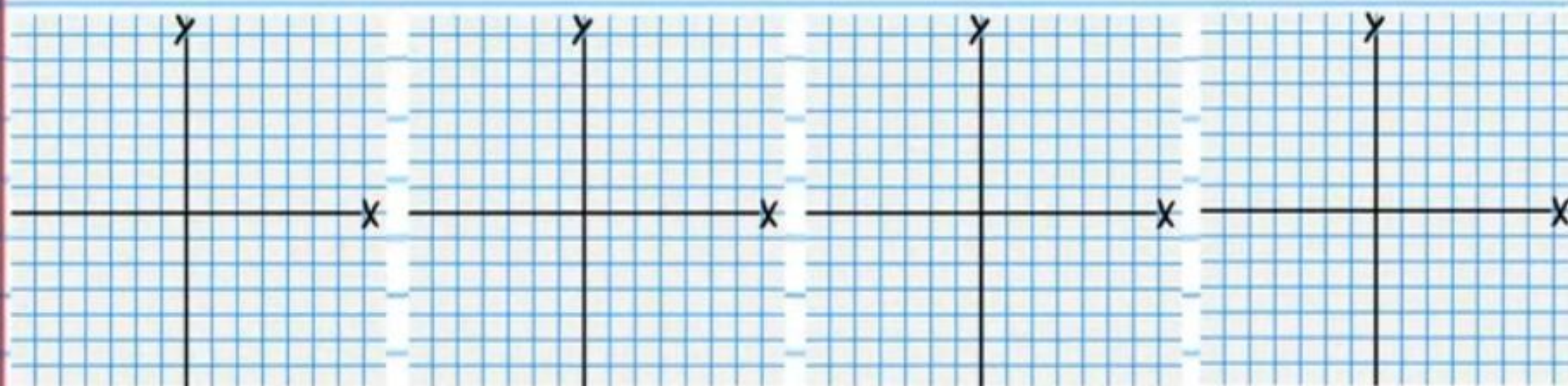
Parent Relation Transformation #1 Transformation #2 Final Transformation



⑨ $y = 2(3^{-x-1}) - 1$

Altered Form:

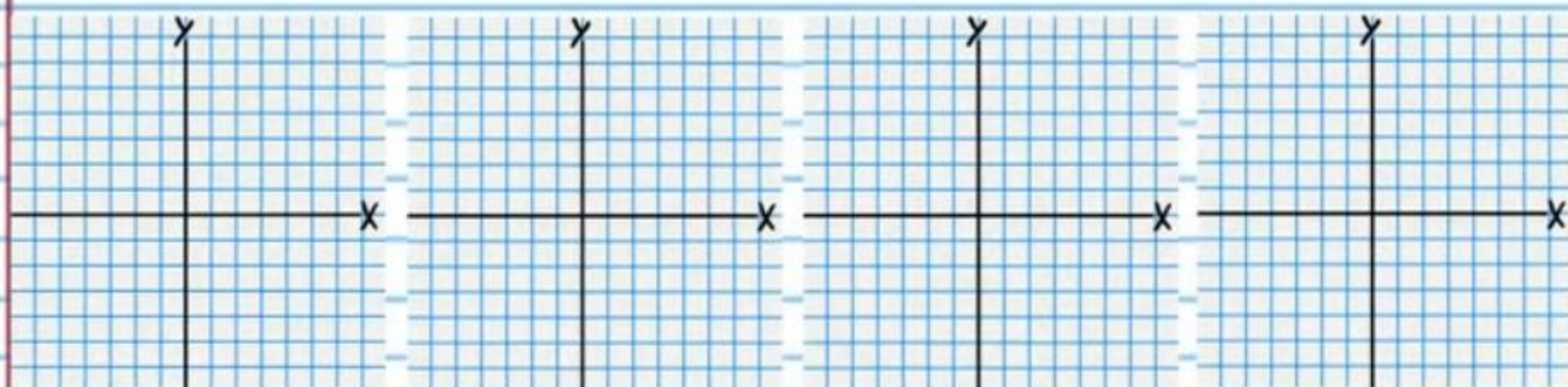
Parent Relation Transformation #1 Transformation #2 Final Transformation



⑩ $y = -3\sqrt{x+2} + 3$

Altered Form:

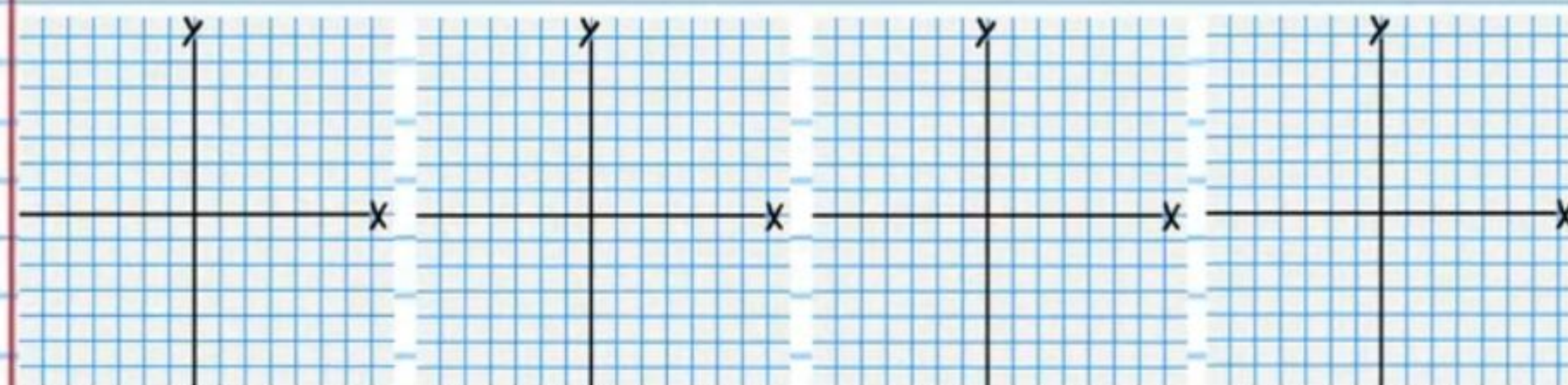
Parent Relation Transformation #1 Transformation #2 Final Transformation



⑪ $y = -\frac{2^{x+2}}{3} + 1$

Altered Form:

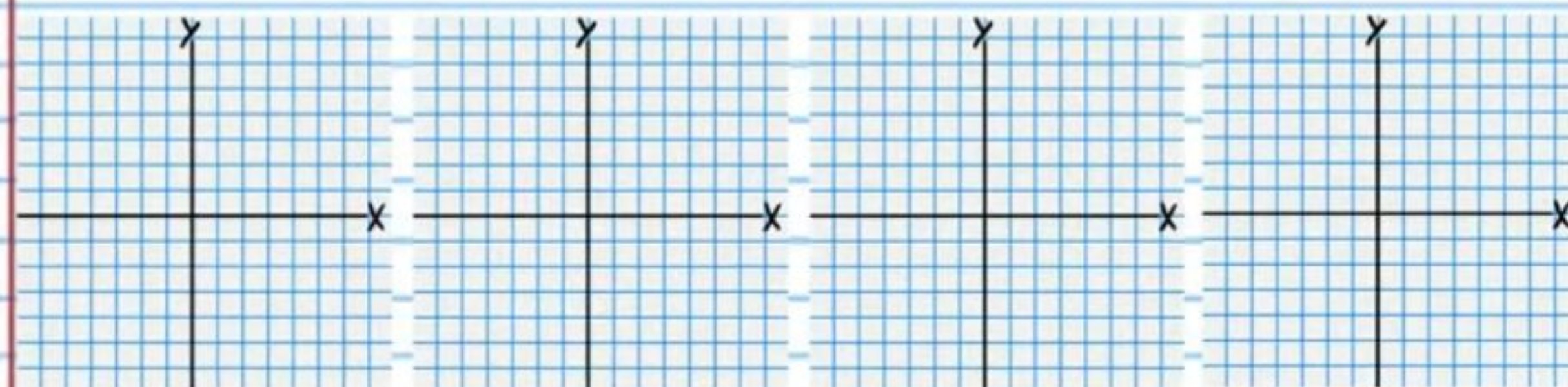
Parent Relation Transformation #1 Transformation #2 Final Transformation



⑫ $y = 3 - (4x+2)^3$

Altered Form:

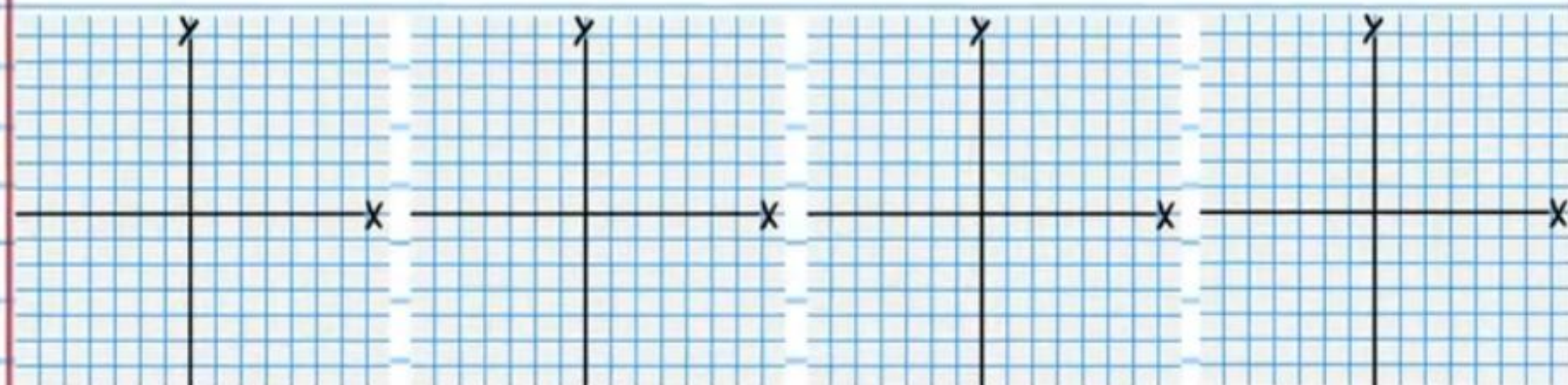
Parent Relation Transformation #1 Transformation #2 Final Transformation



⑬ $y = -\frac{4}{x+1} + 3$

Altered Form:

Parent Relation Transformation #1 Transformation #2 Final Transformation

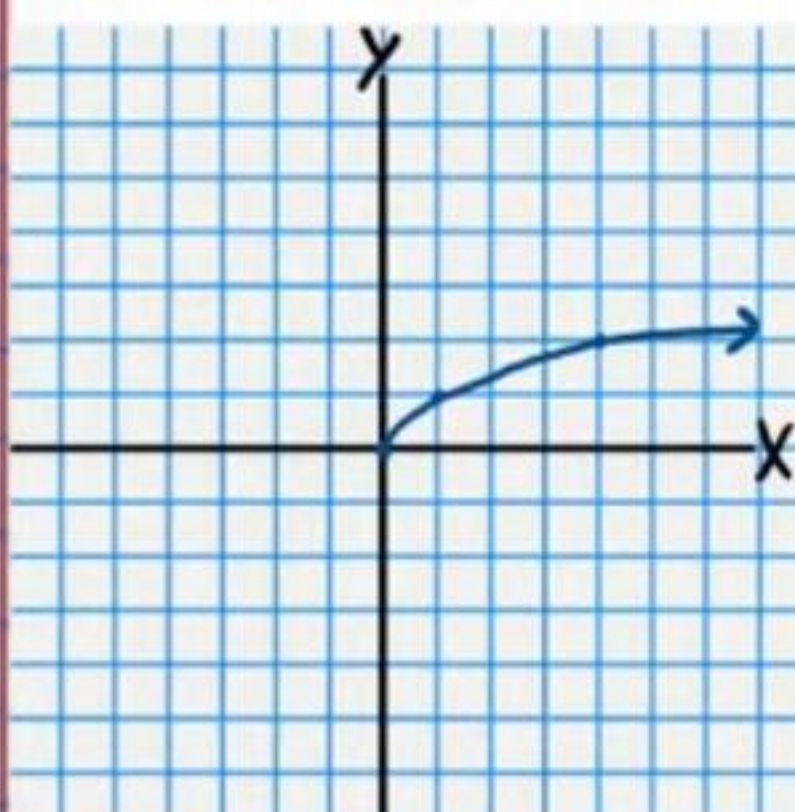


Answers

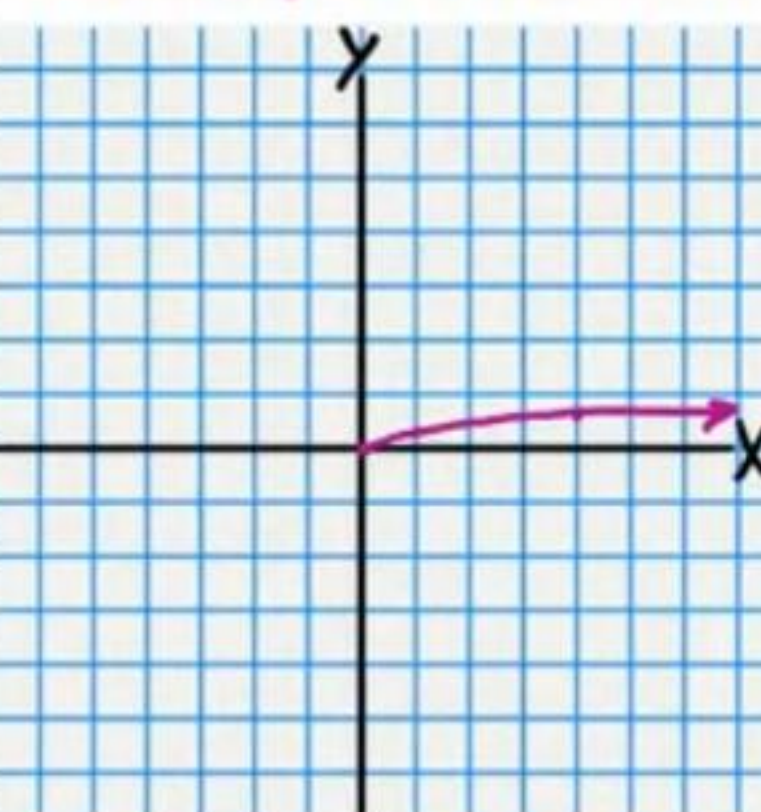
① $y = \frac{\sqrt{x}}{3} - 1$

Altered Form: $3(y+1) = \sqrt{x}$

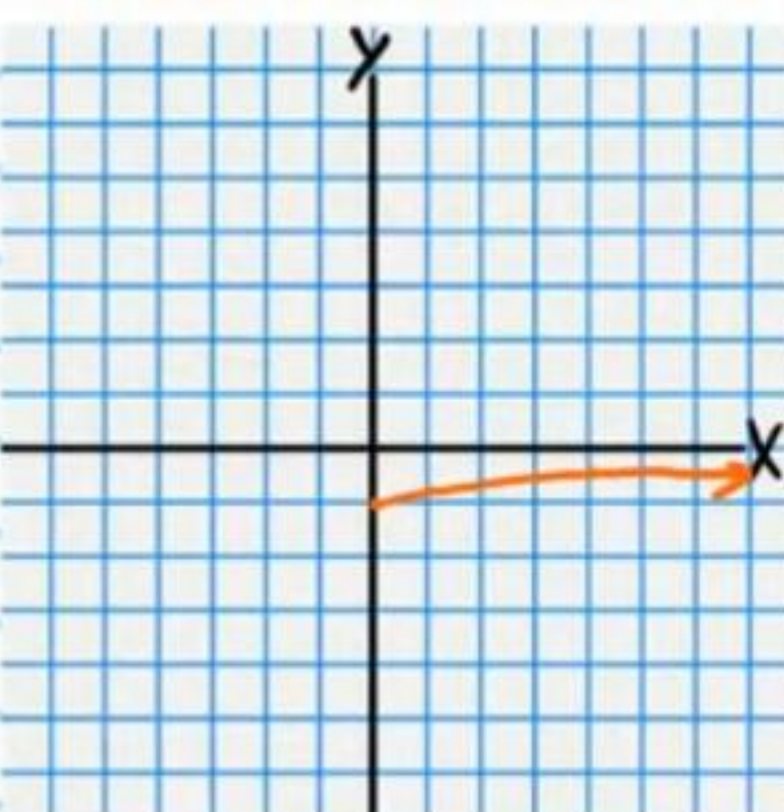
Parent Relation
 $y = \sqrt{x}$



Transformation #1
 $3y = \sqrt{x}$



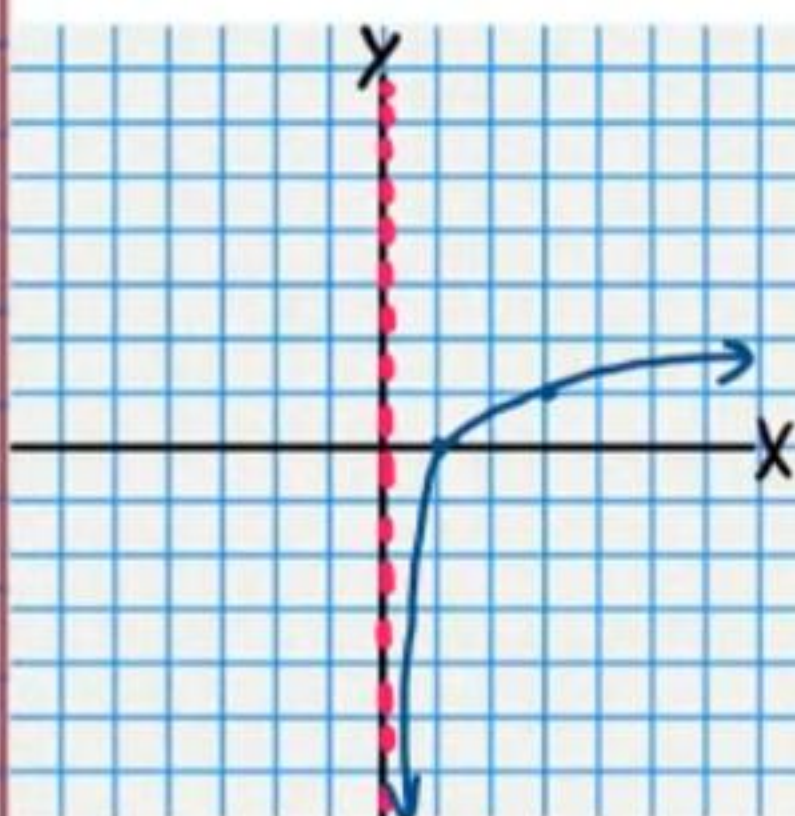
Transformation #2
 $3(y+1) = \sqrt{x}$



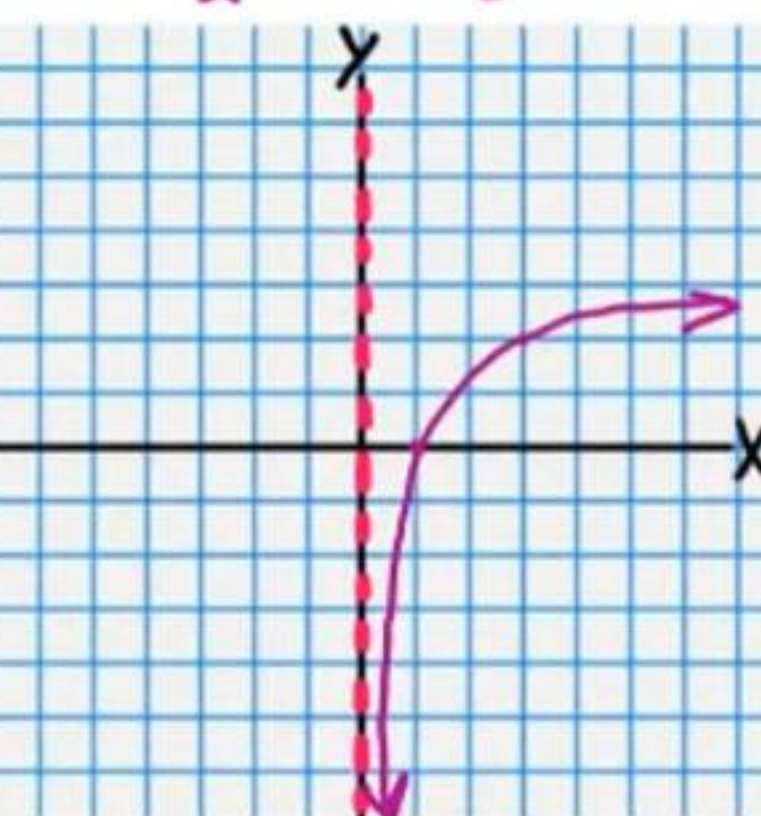
② $y = 2\log_3 x + 2$

Altered Form: $\frac{y-2}{2} = \log_3 x$

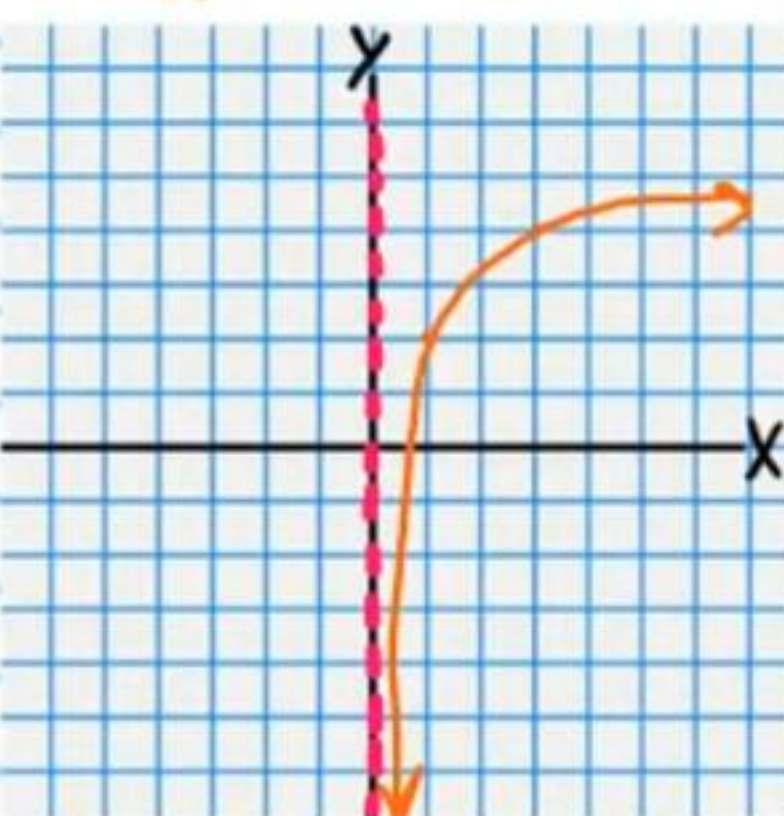
Parent Relation
 $y = \log_3 x$



Transformation #1
 $\frac{y}{2} = \log_3 x$



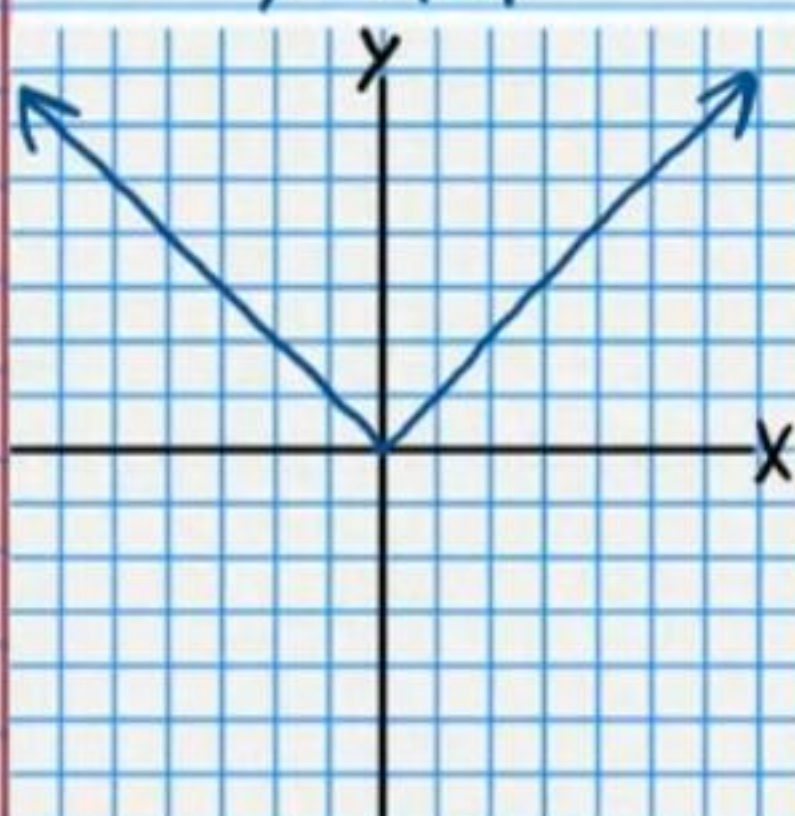
Transformation #2
 $\frac{y-2}{2} = \log_3 x$



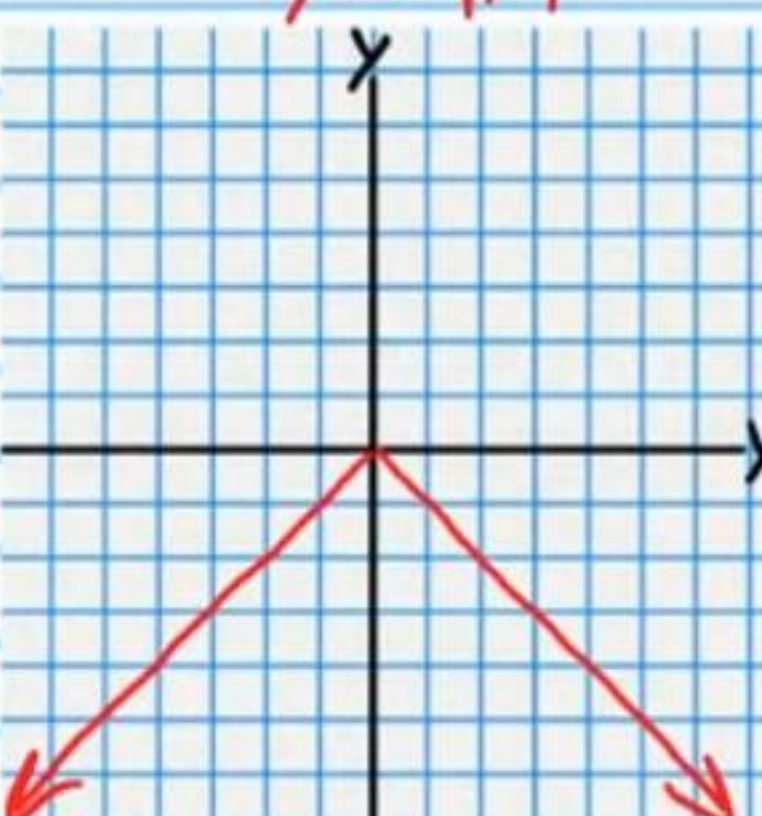
③ $y = -|x| + 5$

Altered Form: $-(y-5) = |x|$

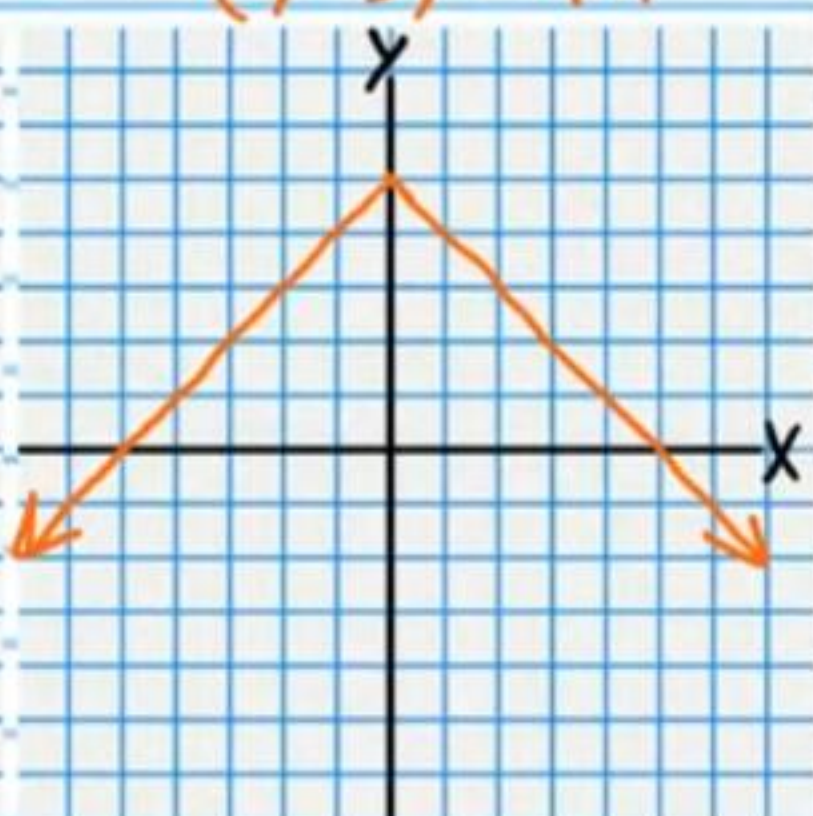
Parent Relation
 $y = |x|$



Transformation #1
 $-y = |x|$



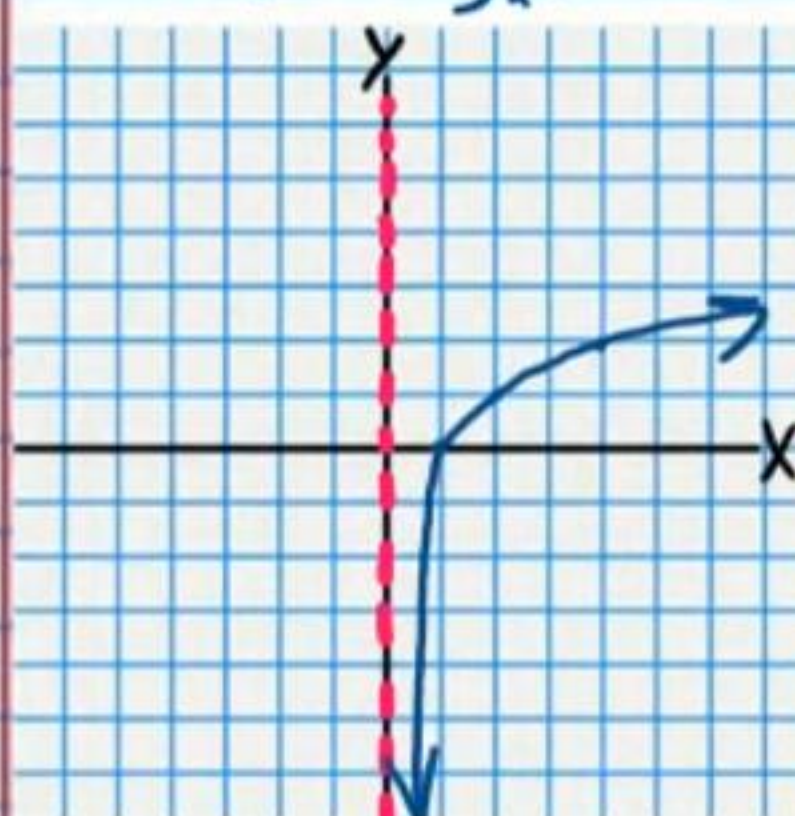
Transformation #2
 $-(y-5) = |x|$



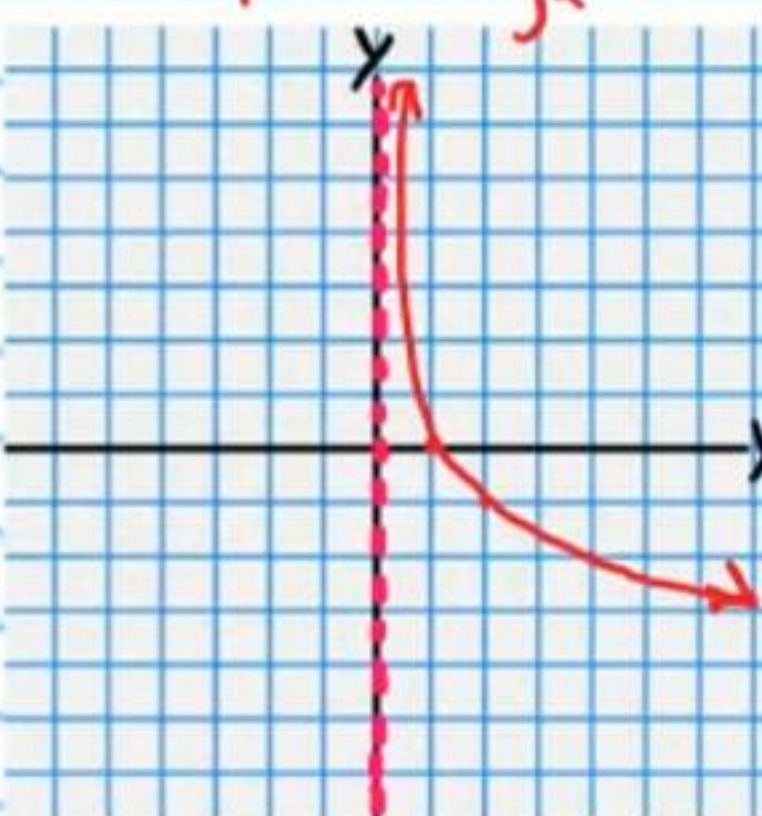
④ $y = -\log_2 x - 3$

Altered Form: $-(y+3) = \log_2 x$

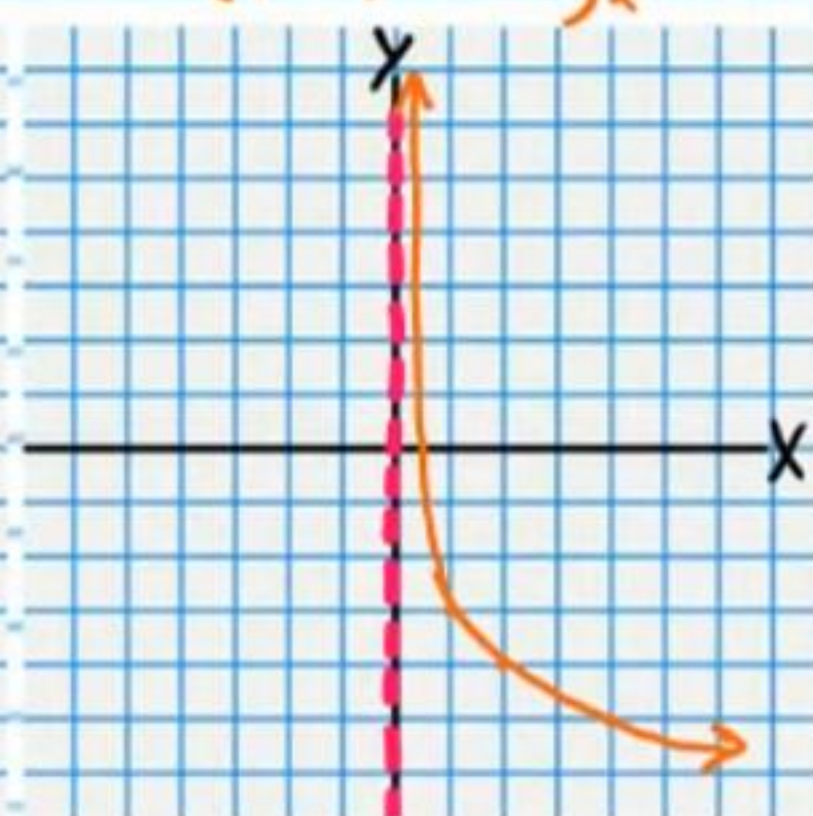
Parent Relation
 $y = \log_2 x$



Transformation #1
 $-y = \log_2 x$



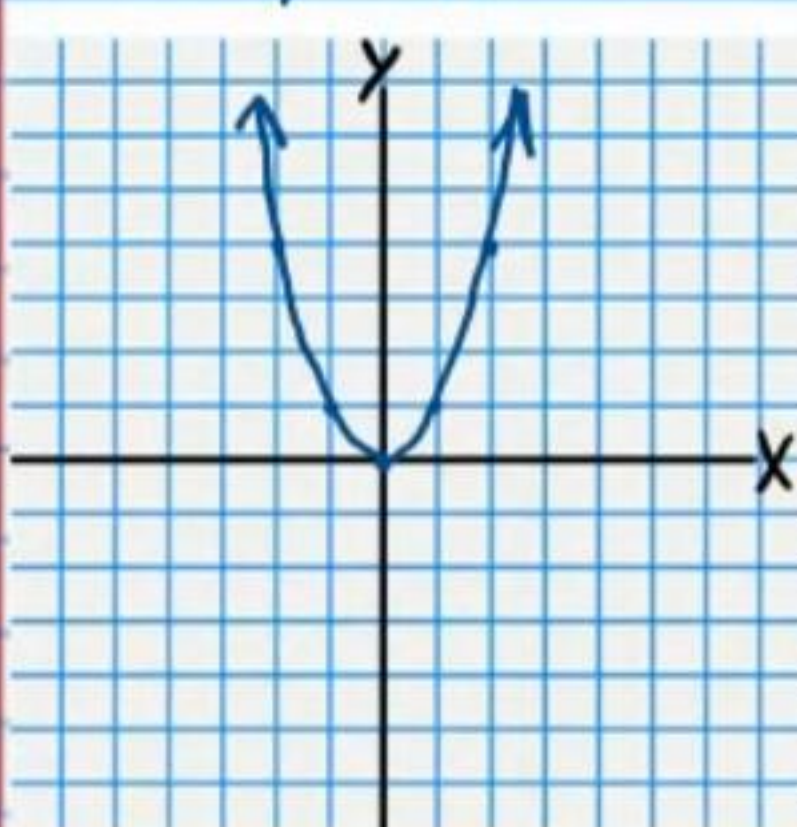
Transformation #2
 $-(y+3) = \log_2 x$



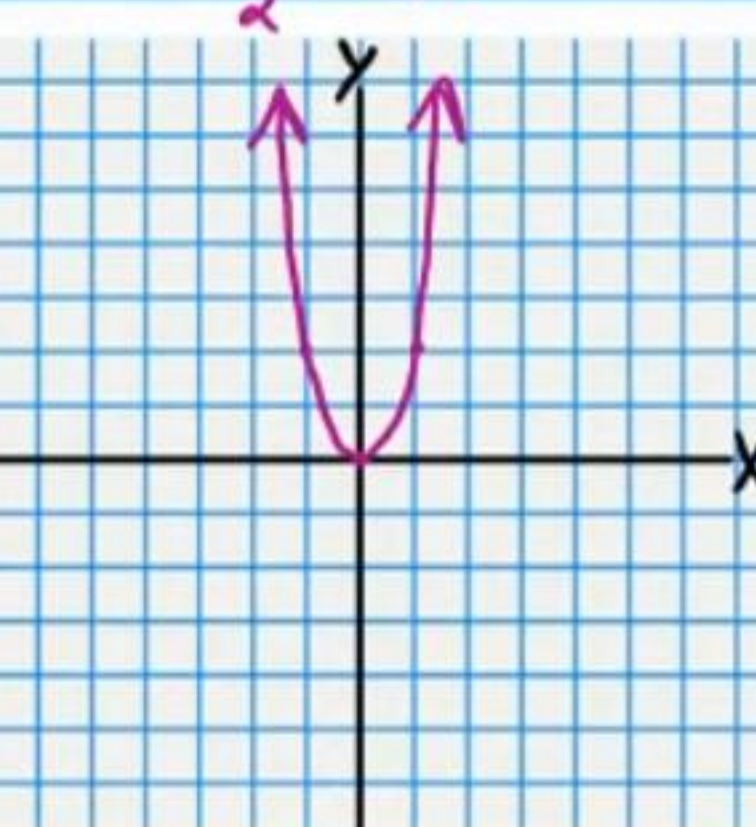
⑤ $y = 2(x+1)^2 - 4$

Altered Form: $\frac{y+4}{2} = (x+1)^2$

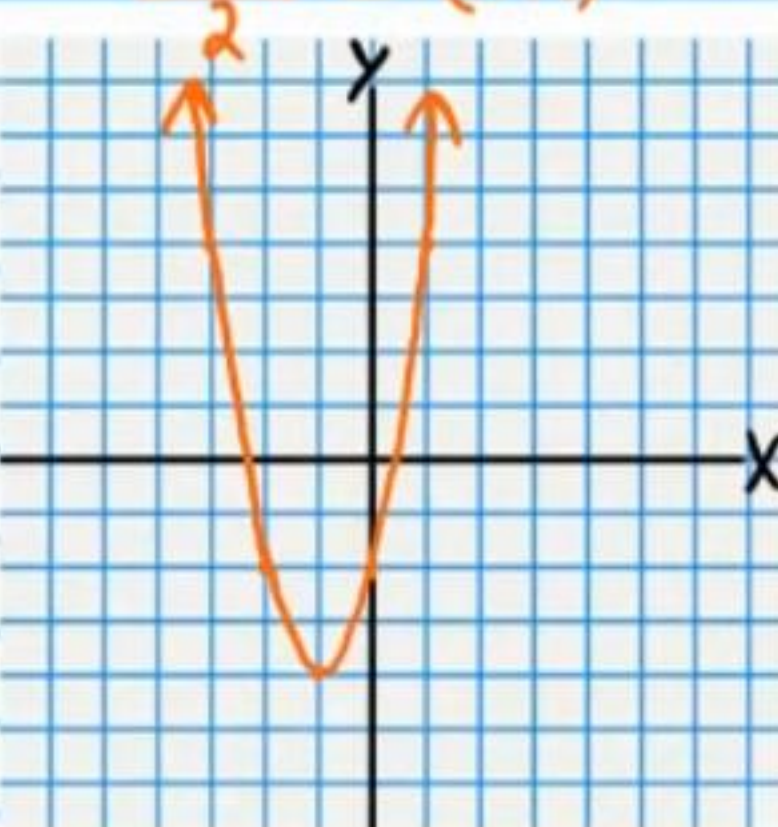
Parent Relation
 $y = x^2$



Transformation #1
 $\frac{y}{2} = x^2$



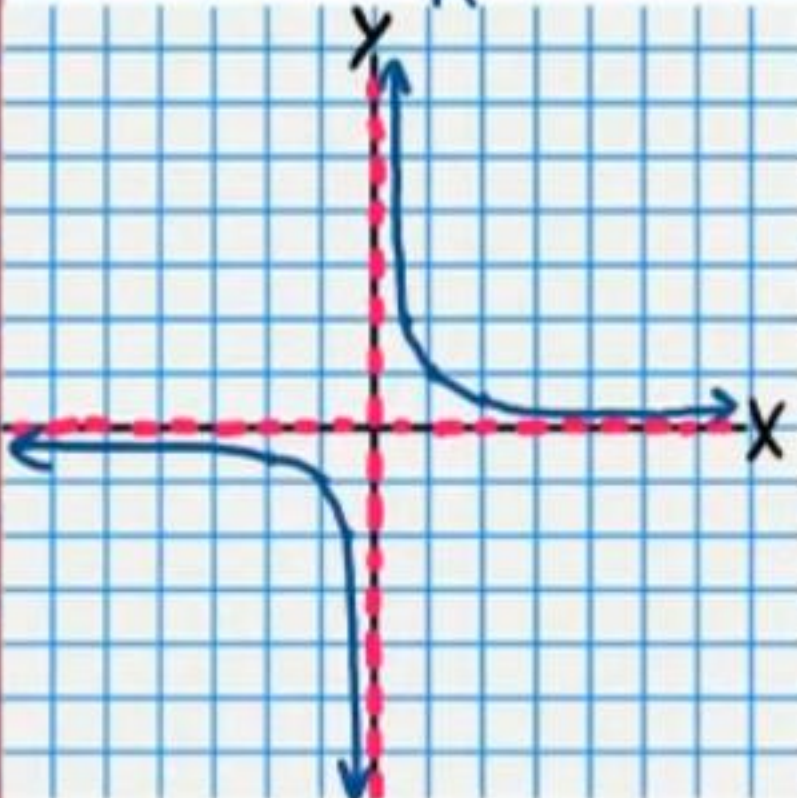
Transformation #2
 $\frac{y+4}{2} = (x+1)^2$



⑥ $y = -\frac{1}{x-2} + 1$

Altered Form: $-(y-1) = \frac{1}{x-2}$

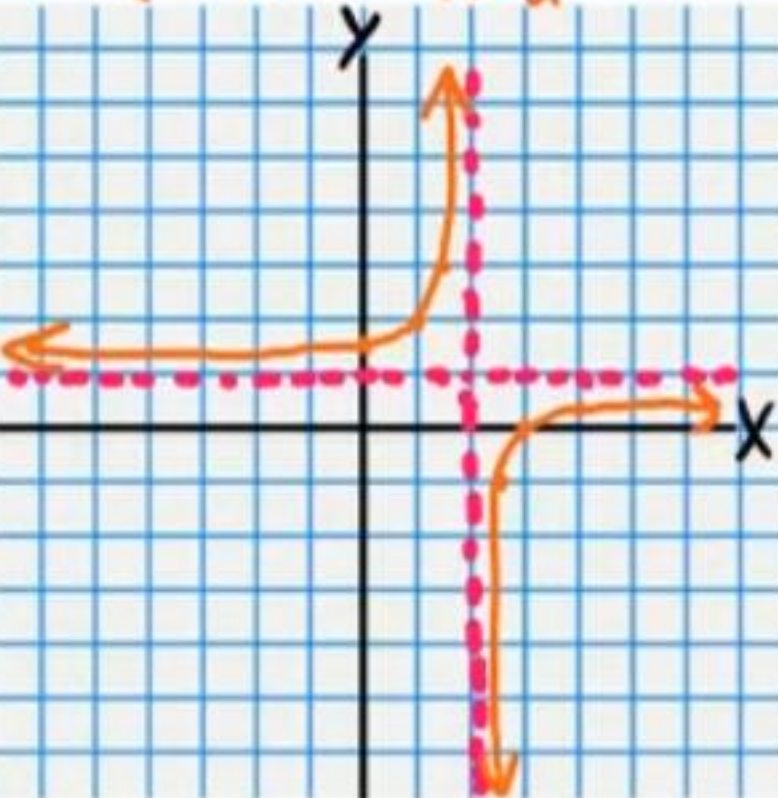
Parent Relation
 $y = \frac{1}{x}$



Transformation #1
 $-y = \frac{1}{x}$



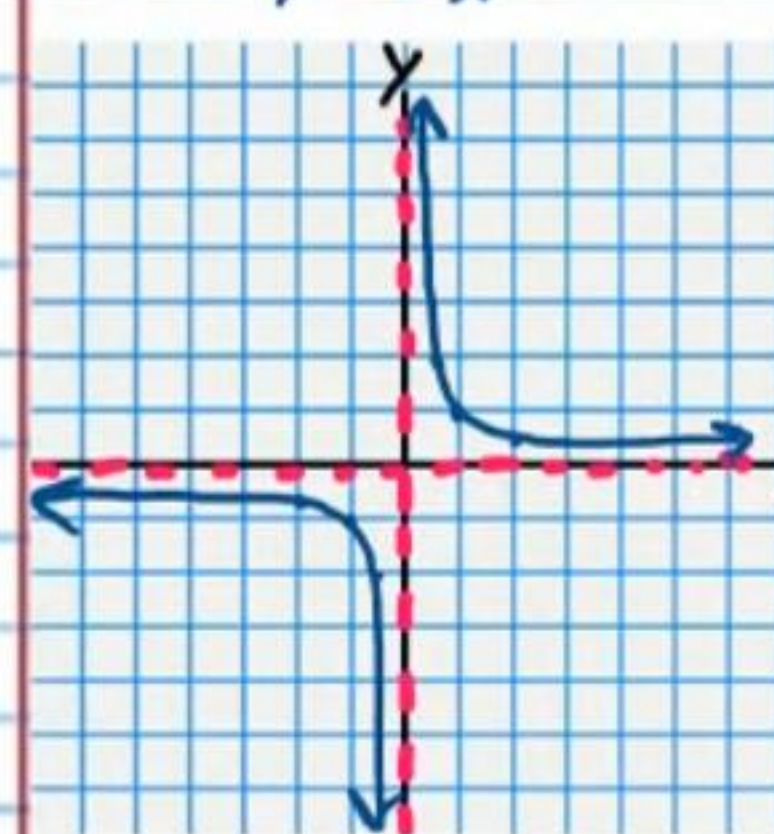
Transformation #2
 $-(y-1) = \frac{1}{x-2}$



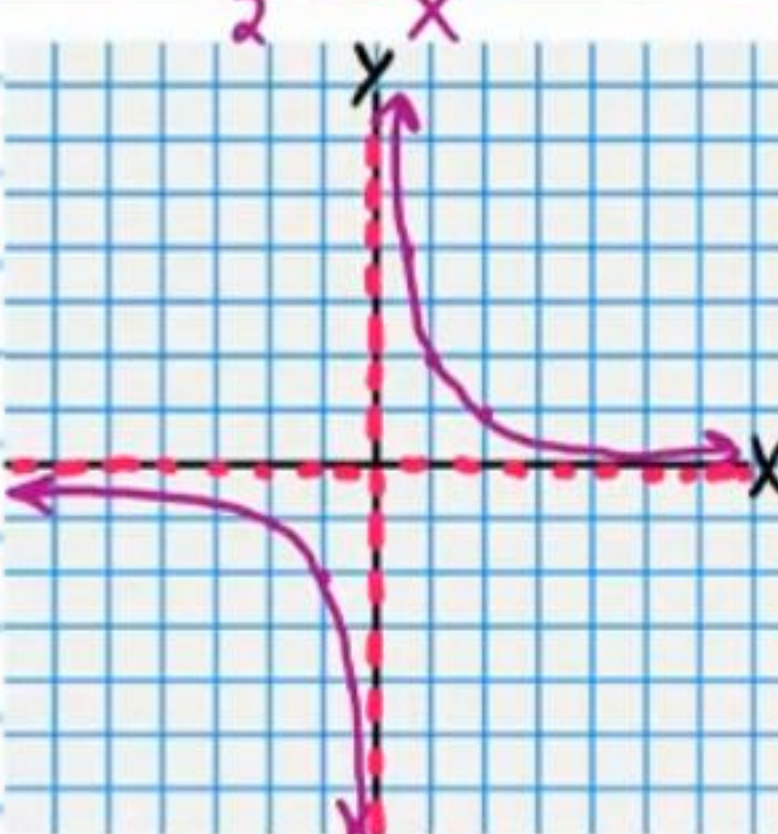
⑦ $y = \frac{2}{x+3} - 3$

Altered Form: $\frac{y+3}{2} = \frac{1}{x+3}$

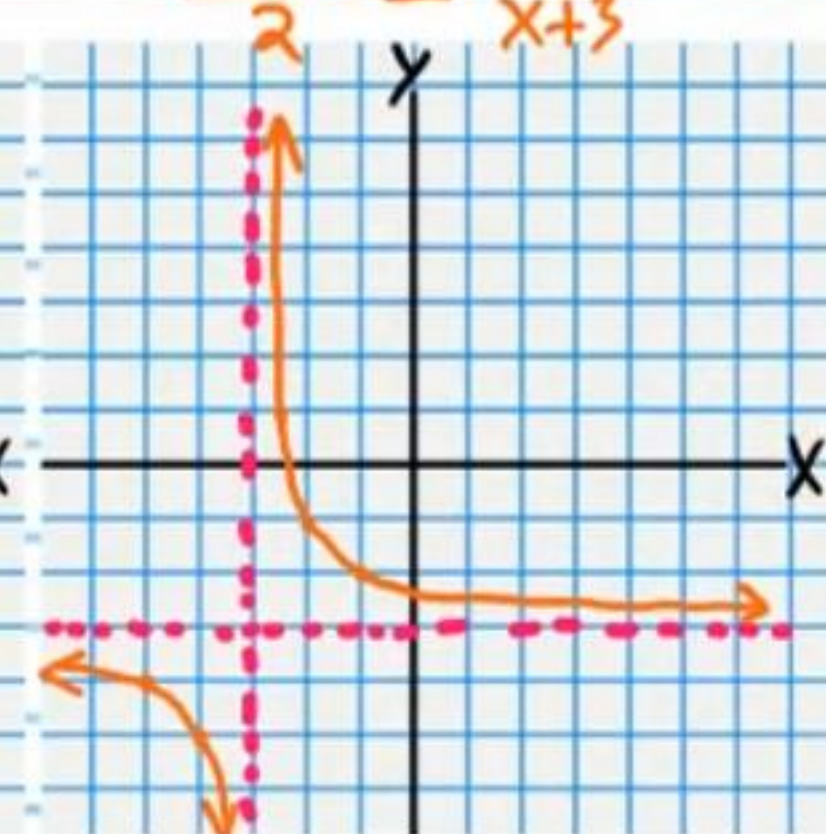
Parent Relation
 $y = \frac{1}{x}$



Transformation #1
 $\frac{y}{2} = \frac{1}{x}$



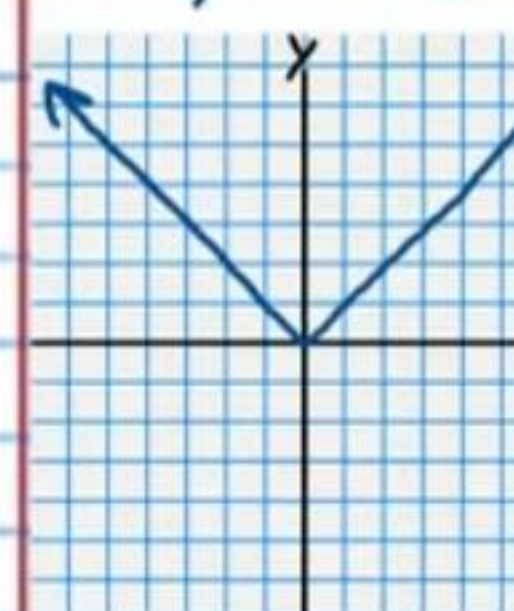
Transformation #2
 $\frac{y+3}{2} = \frac{1}{x+3}$



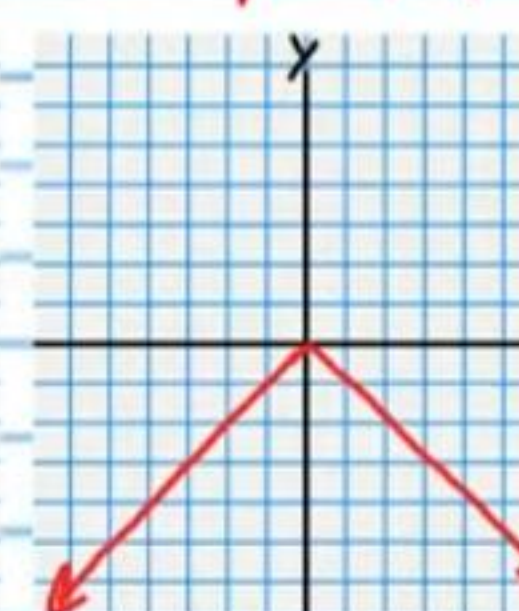
⑧ $y = -\left|\frac{x}{2} - 3\right| + 3$

Altered Form: $-(y-3) = \left|\frac{x}{2} - 3\right|$

Parent Relation
 $y = |x|$



Transformation #1
 $-y = |x|$



Transformation #2
 $-(y-3) = |x-3|$

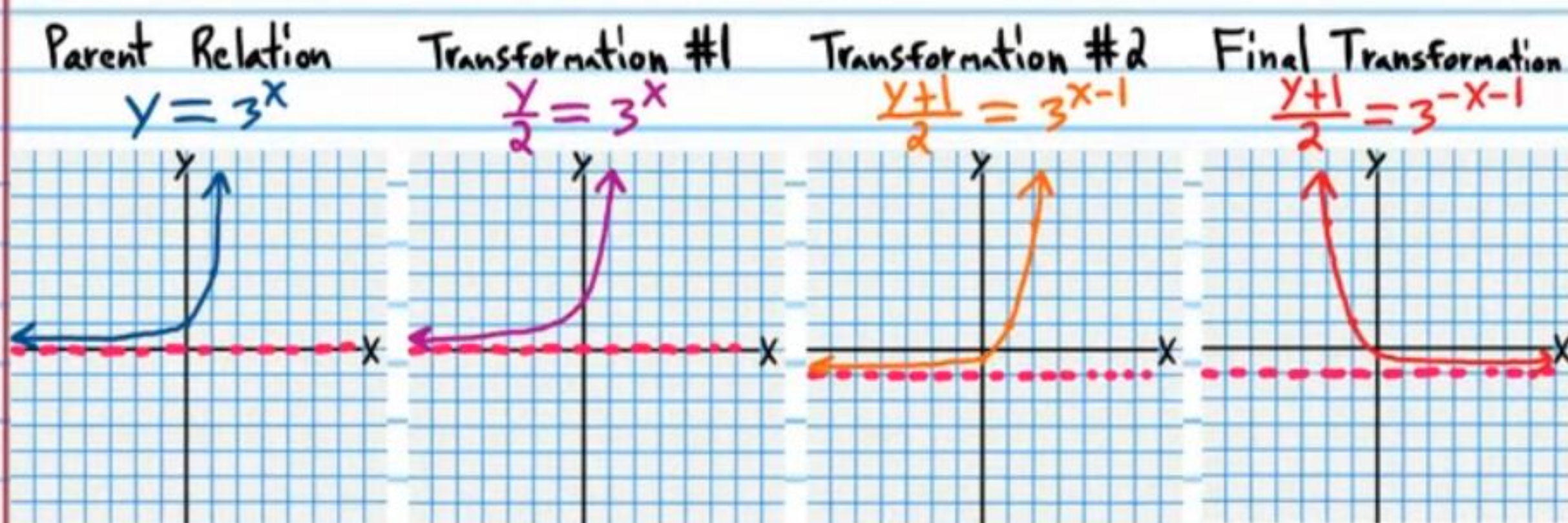


Final Transformation
 $-(y-3) = \left|\frac{x}{2} - 3\right|$



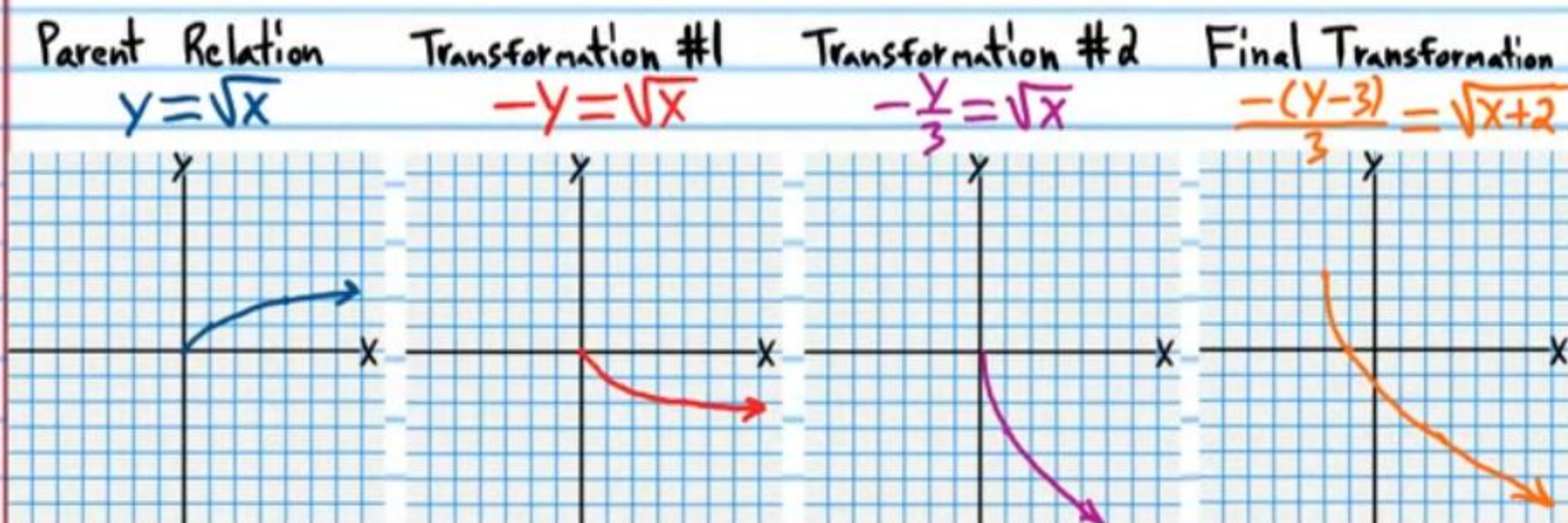
⑨ $y = 2(3^{-x-1}) - 1$

Altered Form: $\frac{y+1}{2} = 3^{-x-1}$



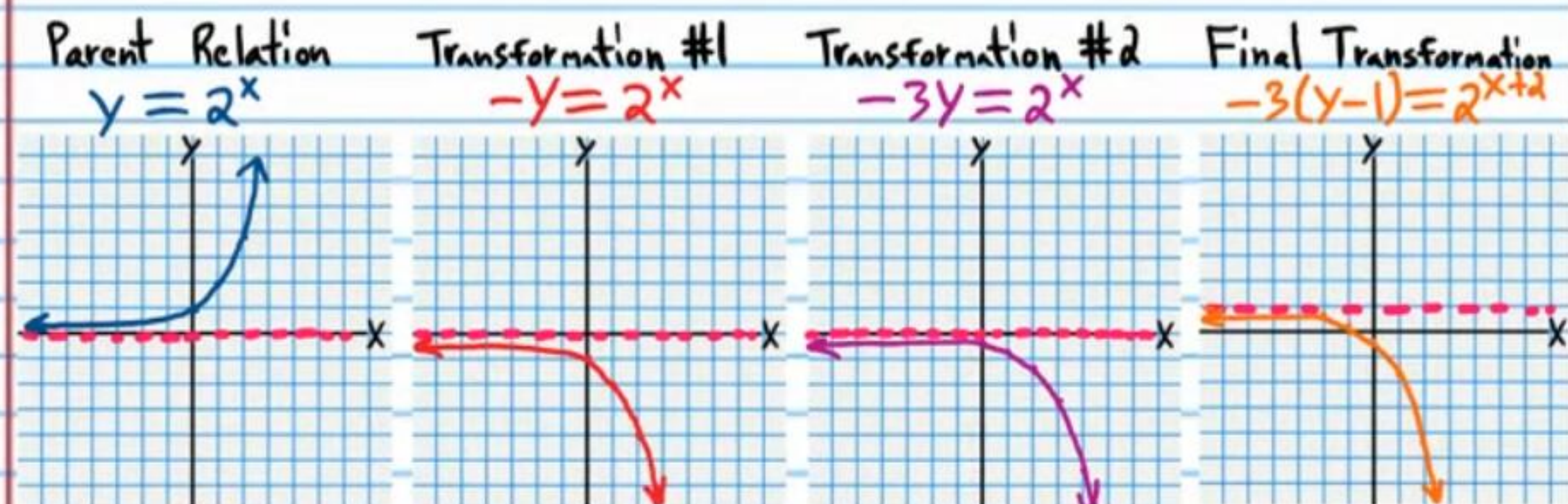
⑩ $y = -3\sqrt{x+2} + 3$

Altered Form: $\frac{-(y-3)}{3} = \sqrt{x+2}$



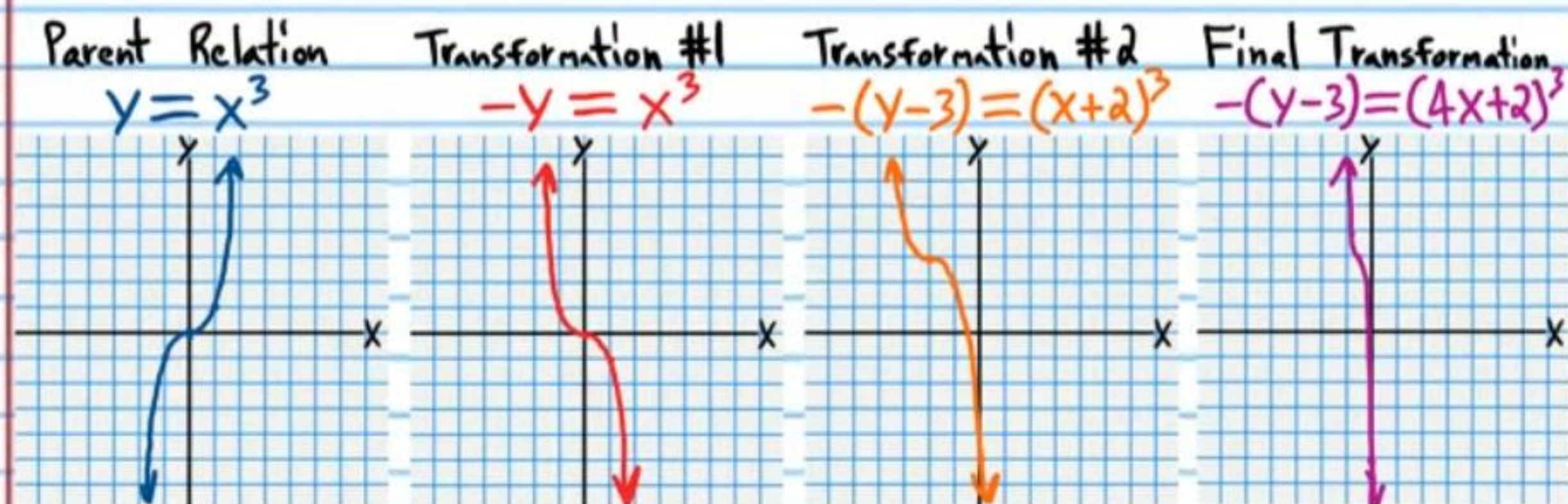
⑪ $y = -\frac{2^{x+2}}{3} + 1$

Altered Form: $-3(y-1) = 2^{x+2}$



⑫ $y = 3 - (4x+2)^3$

Altered Form: $-(y-3) = (4x+2)^3$



⑬ $y = -\frac{4}{x+1} + 3$

Altered Form: $-\frac{y-3}{4} = \frac{1}{x+1}$ or $\frac{-(y-3)}{4} = \frac{1}{x+1}$

